Introduction to Neurological Conditions & Chronic Stroke

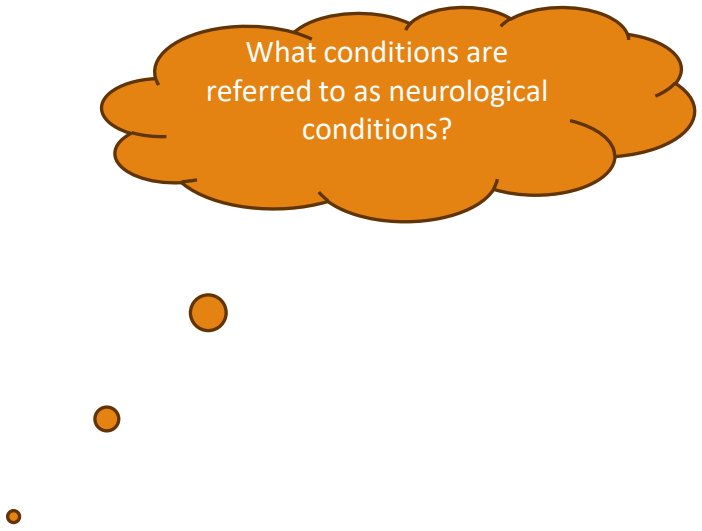
PROF ISAAC SORINOLA

Learning Outcomes

By the end of the session, students should be able to:

- Identify possible domains of health affected in clients with neurological conditions
- Explain common signs and symptoms associated with neurological conditions
- Identify associated psychological and social problems associated with neurological conditions
- Use ICF to characterise a common neurological condition – chronic stroke.

Neurological conditions



Disorders affecting the nervous system

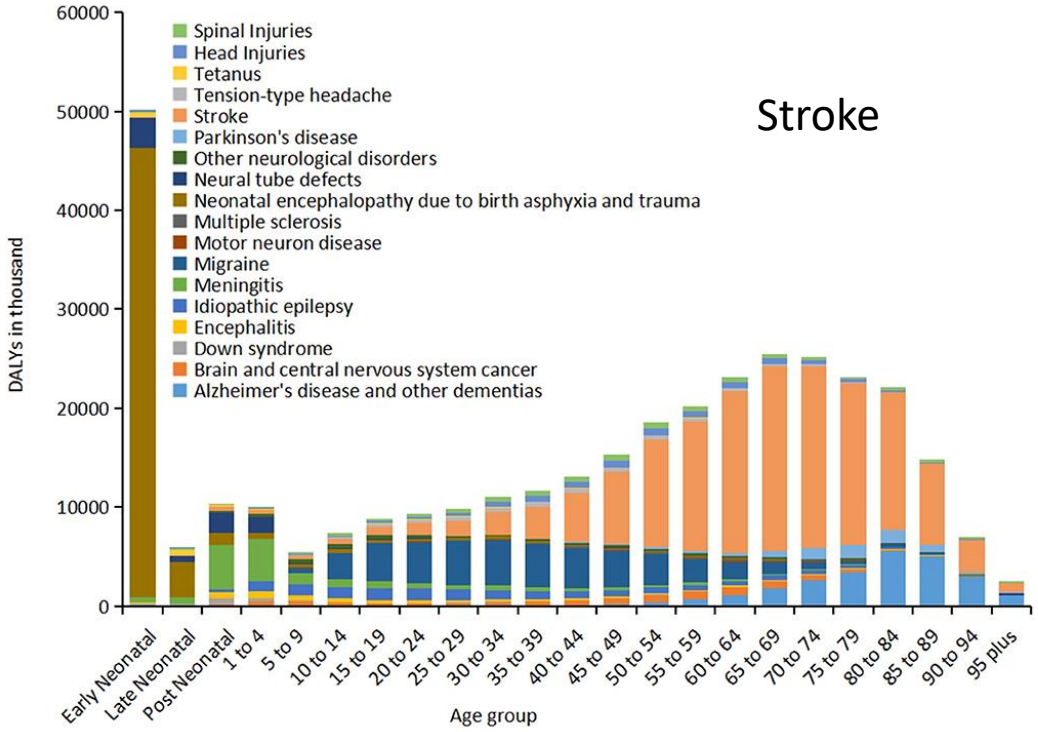
- CNS: Brain and spinal cord
- PNS: peripheral nerves and muscles

Major and increasing global health challenge

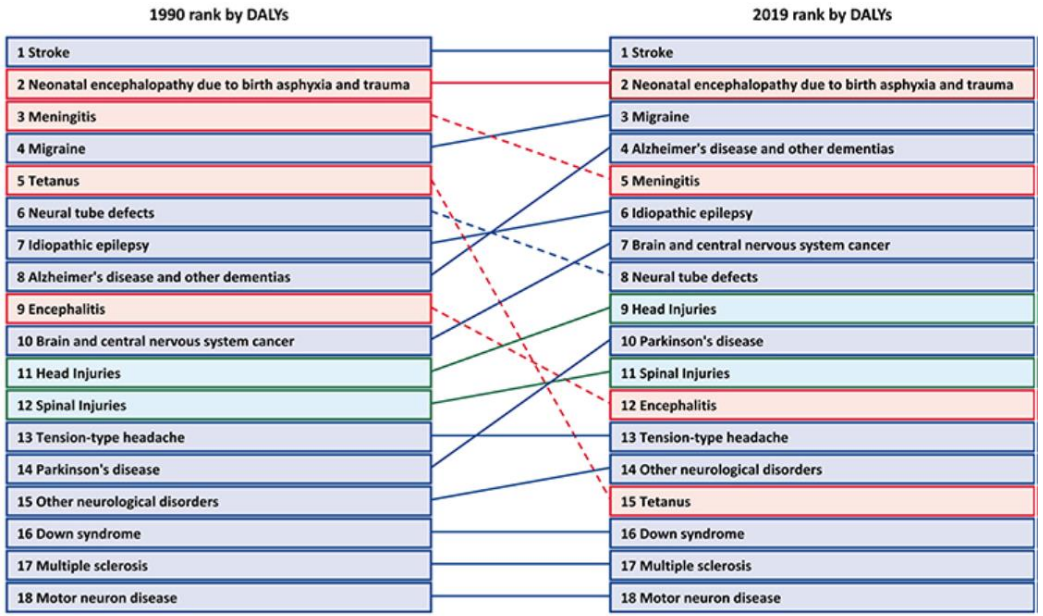
Impairs cognitive-motor function

Ageing adds to the prevalence of neurological conditions

Burden of neurological conditions



Stroke



Disability-adjusted life year (DALY) - measure of overall disease burden

Client groups

Age groups

Perinatal

Infancy

Childhood

Adolescence

Adulthood

Elderly

Frail elderly

Pathologies

Genetic

Acquired

Developmental

Degenerative

Acute illness

Sign & Symptoms

- Arousal
- Motor
- Sensory
- Cognitive & perception
- Psychological
- Functional ability

Life stages – the impact of neurological conditions

Baby/Toddler – development / exploration / play

Child - development / play / learning

Adolescent – development / learning / leisure / social

Adult/Middle age - employment / family / self-care

Older adult/Old age – self-care / mobility / independence



Aetiology of neurological conditions

Sudden onset conditions – acute onset, followed by partial recovery e.g., acquired brain injury such as stroke or spinal cord injury

Intermittent and unpredictable conditions - where relapses and remissions lead to marked variation in the care needed. e.g., epilepsy, certain types of headache or early multiple sclerosis

Progressive conditions - where progressive deterioration in neurological function leads to increasing dependence on help and care from others. e.g., Motor Neurone Disease (MND), Parkinson's Disease (PD).

Stable neurological conditions - but with changing needs due to development or ageing. E.g., post-polio syndrome or cerebral palsy in adults.

Organisation of the Nervous system

Nervous System

CNS
Brain
Spinal cord

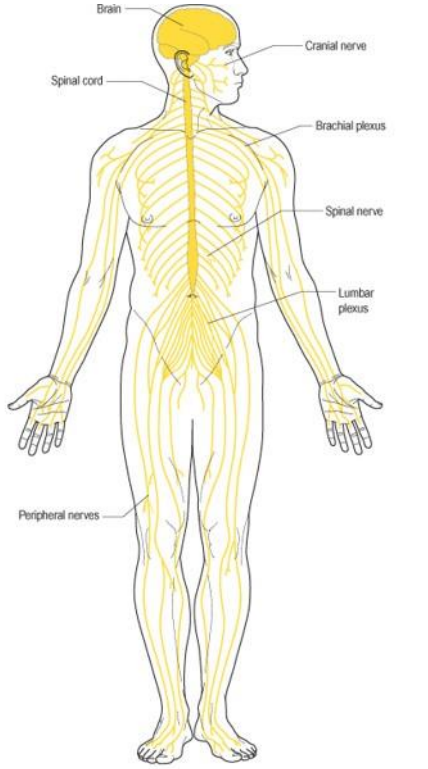
PNS

Peripheral nerves

Plexuses

Sensory receptors

Ganglia



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Common causes of CNS Lesions

Extrinsic disorders
- Compression of the brain and SC
  - Haematomas, abscess, tumours, trauma
  - Hydrocephalus

Vascular disorders e.g. stroke
- Thrombosis
- Haemorrhage

Systemic disorders
- Metabolic
- Toxic e.g. alcohol, drugs
- Nutritional
- Immunological – SLE, sarcoidosis

Intrinsic disorders
- Metabolic
- Infective e.g. meningitis
- Neoplastic – 1^0, 2^0 or paraneoplastic
- Degenerative disorders
  - Parkinson's disease
- Paroxysmal disorders
- Immunological
- Genetic

Common signs associated with CNS lesions

Motor control problems

Abnormal movements
- Athetosis
- Chorea,
- Hemiballismus
- Dystonia
- Abnormal movement pattern

Sensory problems
- Loss of sensations
- Abnormal sensations

Perceptual deficits
- Unilateral neglect

Cognitive deficits
- attention deficits, short term memory loss, forgetfulness etc

Behavioural problems
- such as excessive laughing or anger

Communication problems
- Aphasia, (global, sensory or motor)

Visual problems
- Cortical blindness, inability to recognise people etc
- Hemianopsia

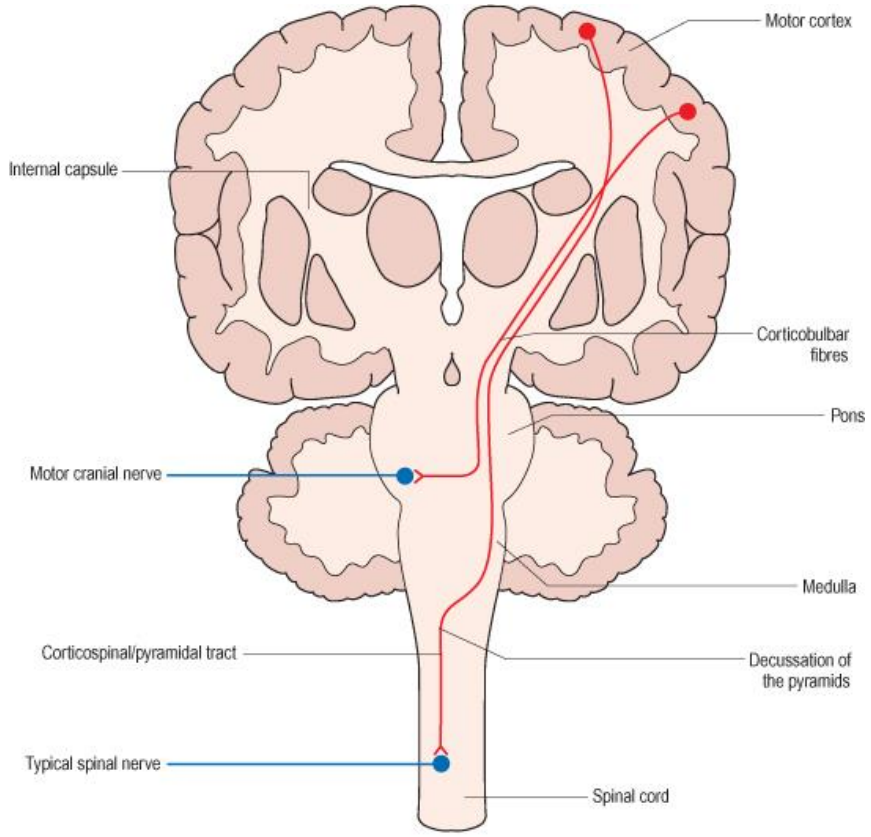
Motor control problems

Upper motor neuron (CNS)

Hypertonicity/Spasticity

- increased resistance to passive stretching of muscles

- No atrophy
  - wasting of muscles
- Hyperactive reflexes
  - hyperreflexia
- Abnormal posture
- Associated reactions
  - Clonus, positive Babinski response
- Loss of dexterity
- Muscle weakness or paralysis
- Increased fatigability



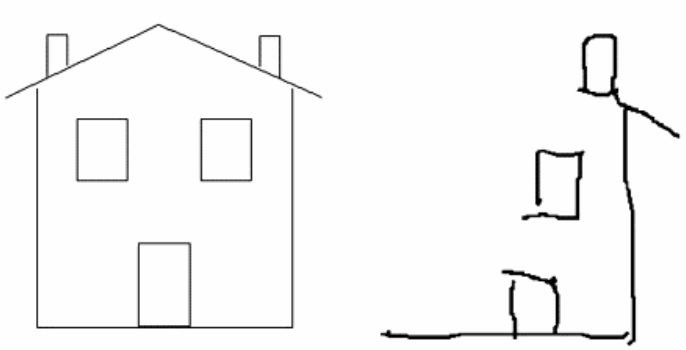
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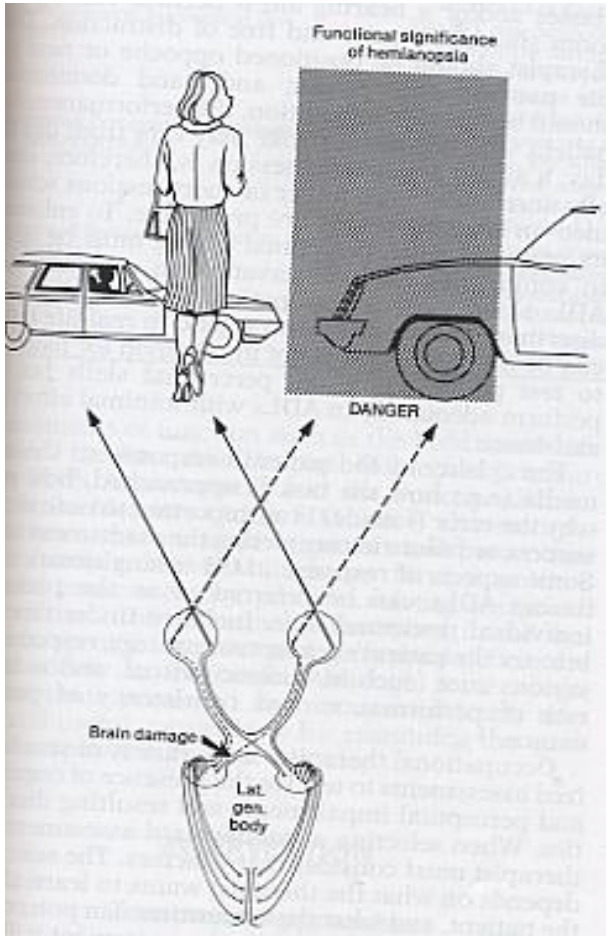
Abnormal movement with CNS lesions





Problems related to CNS lesions





Signs associated with specific area of the CNS

Cerebral Cortex e.g., Stroke	Hemiplegia, spasticity, balance problems, loss of movement and dexterity, muscle weakness, blindness etc
Basal ganglia e.g., Parkinson etc	Rigidity and or spasticity, abnormal movements (chorea, ballismus,akinesia), postural problems, tremor at rest, scanning speech, dystonia
Cerebellum e.g., head injuries etc	Abnormal muscle tone, ataxia, incoordination, slurred speech, intention tremors, dysmetria

Peripheral Nervous System - Overview

Peripheral nerves (cranial and spinal nerves)

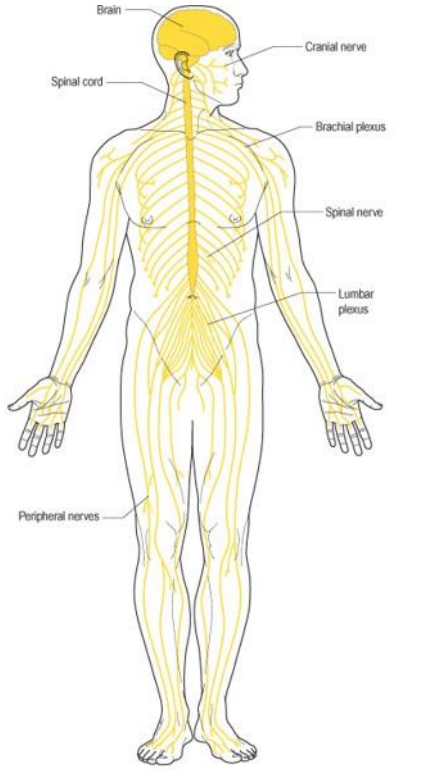
- Cranial nerves (12 pairs)
- 31 pairs of spinal nerves

Plexuses

Ganglia (DRG & autonomic ganglia)

Nerve endings

Muscles



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Common causes of PNS damage

Medical (Neuropathies)

- Systemic diseases
- Drugs and toxins
- Inflammatory
- Nutritional deficiencies
- Hereditary

Injuries

- Open
- Closed

Common signs associated with PNS lesions

α-motor neurone pathology

- **Poliomyelitis** *Spinal Muscular Atrophy, Motor Neurone Disease (MND/ALS)*

Nerve root

Spinal nerve

Plexus Brachial plexus

Peripheral nerve pathology

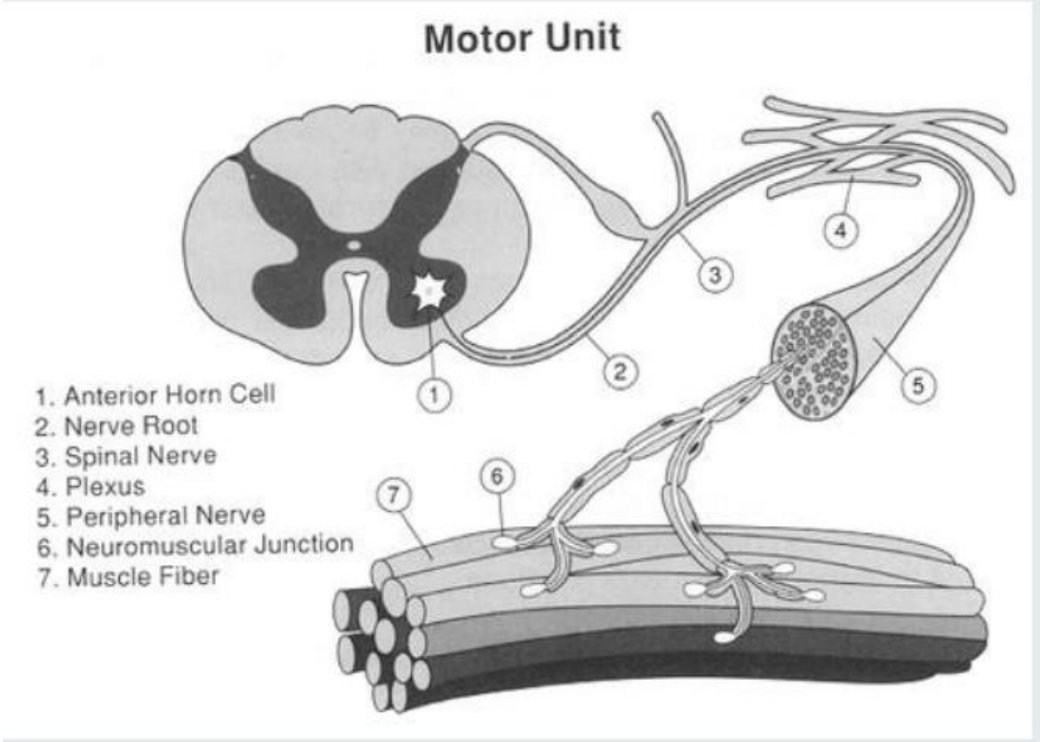
- Peripheral nerve lesion – Common peroneal nerve
- Neuropathy - *Guillain-Barré syndrome, CIDP, Charcot-Marie Tooth disease, Diabetic neuropathy*

Neuromuscular junction disorders

- *Myasthenia Gravis, Lambert Eaton Myasthenic Syndrome*

Muscle pathology or Myopathy

- *Inflammatory (myositis),*
- *Degenerative (dystrophies).*
- *Contractile (myotonias)*



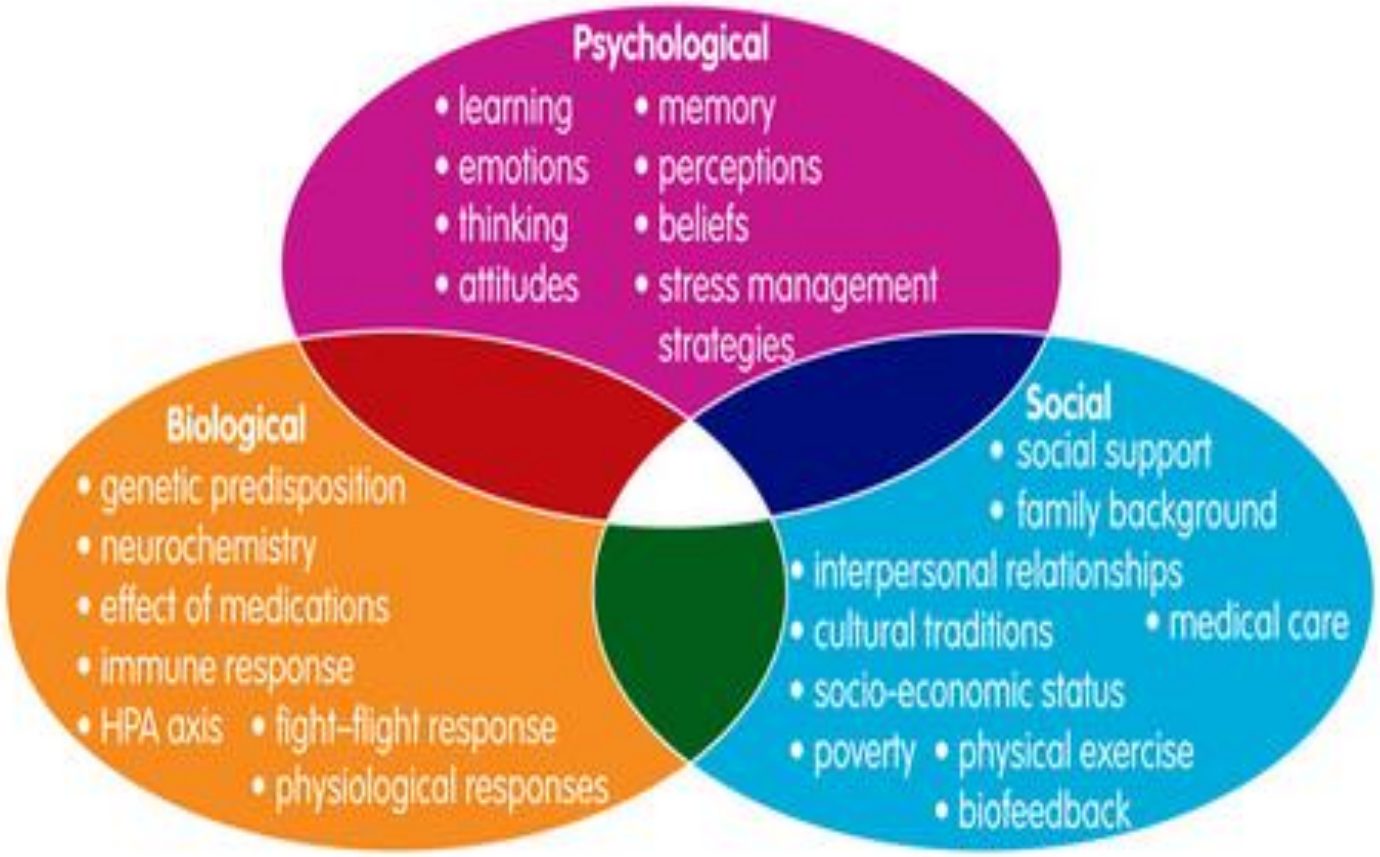
Lower Motor Neuron Lesion

Signs of Lower Motor Neuron

Damaged area	Signs and Symptoms	Example
Anterior Horn Cells	Muscle weakness, flaccid paralysis, deformity	Poliomyelitis
Nerve Roots	Sharp shooting/burning pains Muscle weakness and lost reflexes, deformity	Radiculopathy
Nerve Injuries/nerve damage	Muscle weakness, flaccid paralysis, numbness – loss of sensations (feeling), deformity	Nerve damage, polyneuropathy
Muscle	Muscle weakness with no loss of feeling	Duchenne Muscle Dystrophy

Biopsychosocial framework

- **biopsychosocial framework:** an approach to describing and explaining how *biological*, *psychological* and *social* factors combine and interact to influence physical and mental health



Identify common psychological problems associated with neurological problems

Nobody has responded yet.

Hang tight! Responses are coming in.

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Psychological symptoms associated with neurological conditions

https://healthtalk.org/motor-neurone-disease-mnd/emotional-lability-depression-and-low-mood-with-mnd

Psychological problems associated with neurological conditions

Anxiety

Loss of confidence

Denial

Depression

Frustration

Loneliness

Anger

Nervousness

Embarrassment

Loss of self

Confusion

Isolation

Mood swings

Personality changes

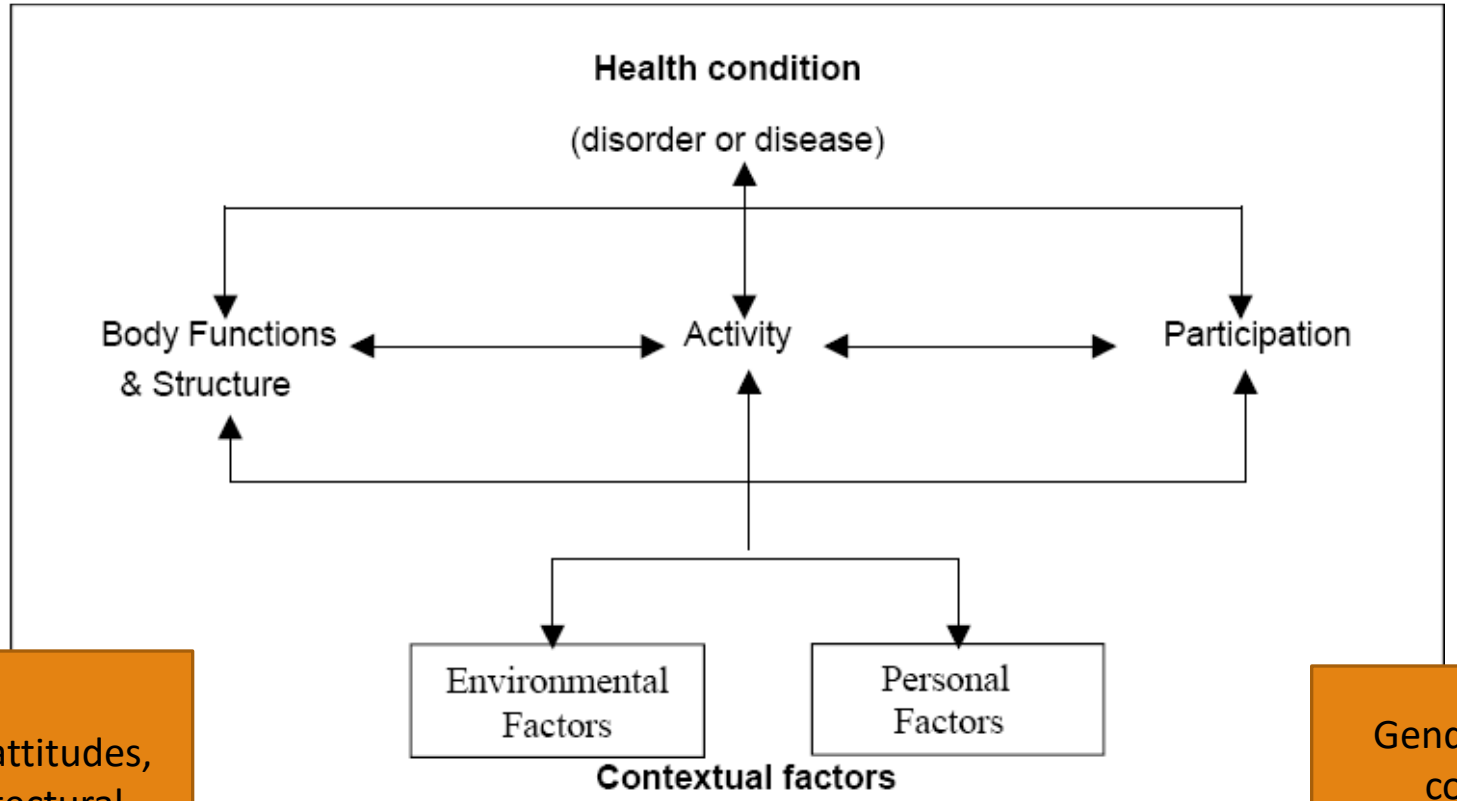
Stress

Demotivated

Common social problems associated with neurological conditions



International Classification of Functioning, Disability and Health (ICF, 2001)



Social attitudes, architectural designs, legal and social structure, climatic terrain

Gender, age, coping strategies, social background, education, past experiences

Definition of Terms in the ICF

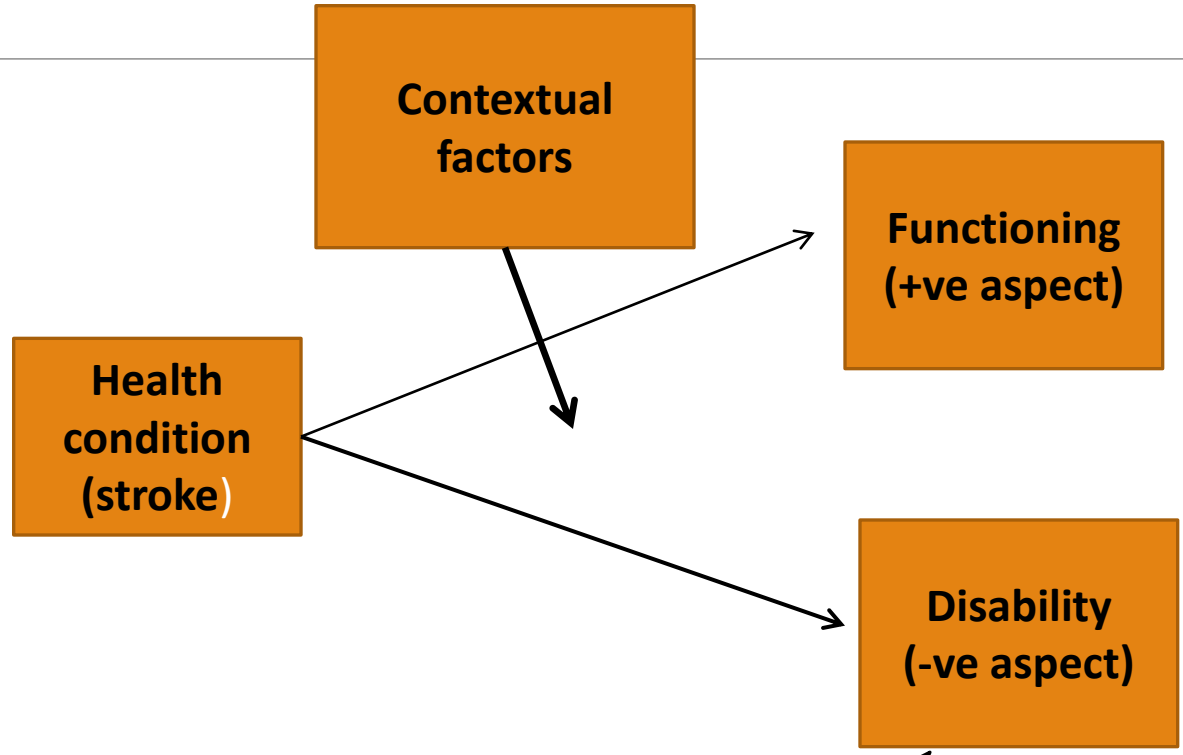
FUNCTIONING - POSITIVE

- Body function refers to the physiological functioning of the body system
- Body structures refers to anatomical parts of the body such as organs limbs etc.
- Activity is the execution of a task or action by an individual e.g., Sit to stand, walking, bed mobility, self-care activities etc.
- Participation in a life situation for example employment, leisure activities etc

DISABILITY – NEGATIVE CONSEQUENCES

- Impairments or problems in body function or structure such as a significant deficit or loss e.g., reduced muscle strength, bulk etc
- Activity limitations are difficulties an individual may have experience in executing activities e.g., Loss of independence in feeding, grooming etc
- Participation restrictions are problems an individual may experience in involvement in life situations e.g., Difficulties encountered accessing employment/university education etc.

International Classification of Functioning, Disability and Health



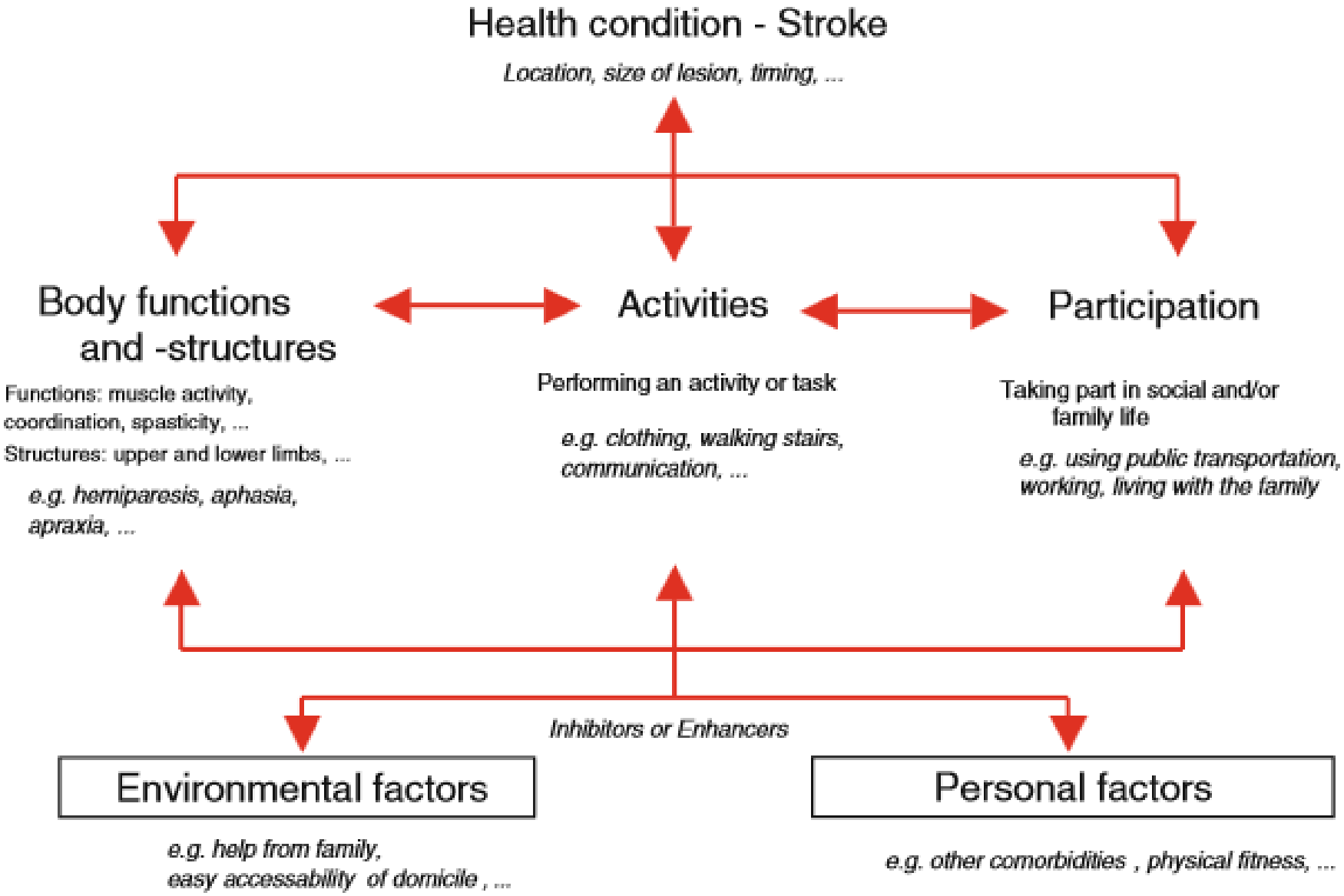
ICF

1. Body functions
2. Body structure
3. Activity
4. Participation

1. Impairment
2. Activity limitations
3. Participation restrictions

Disability is seen as the outcome of the interactions between health conditions and contextual factors

ICF – with Stroke as an example





J Neurol (2012) 259:817–832

Body - domains or areas of function & structure

FUNCTION / PHYSIOLOGY (8)

- Mental/cognitive
- Sensory
- Voice and speech
- Cardiovascular, Haematological and respiratory systems
- Digestive, metabolic, endocrine systems
- Genitourinary & reproductive
- Neuromuscular and movement related
- Skin and related function

STRUCTURE / ANATOMY (7)

- Nervous system
- Eye, ear and related structure
- Cardiovascular, immunological and respiratory system
- Digestive, Metabolic and endocrine system
- Genitourinary and reproductive structure
- Structure related to movement
- Skin and related structures

Body structure and function: Signs and symptoms

- **SIGN** – OBJECTIVE or observable indication of pathological state
  - vital signs - heart rate, blood pressure, brisk reflexes, muscle weakness, colour, temperature etc

- **SYMPTOM** – SUBJECTIVE self-report of underlying pathological state (can only be described by the person experiencing them)
  - e.g. pain, pins and needles in hands, fatigue, anxiety

Activity and Participation domain

Learning and applying knowledge

General task and demands

Communication

Mobility

Self-care

Domestic life

Interpersonal interactions and relationships

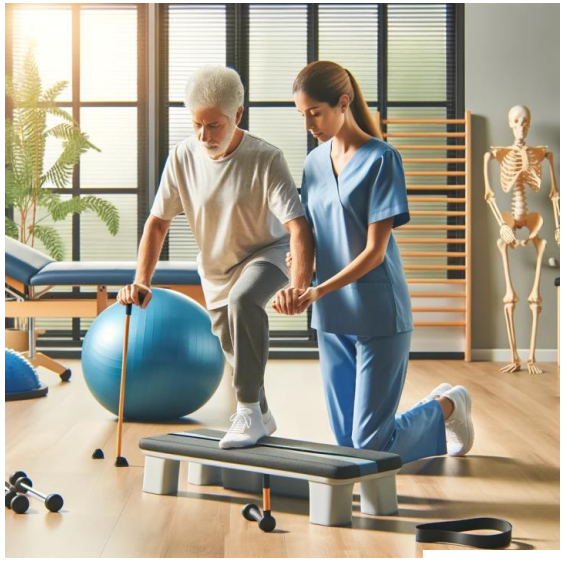
Major life areas: Community, social and civic life

Objectives of Neurorehabilitation

Optimise recovery of physical functioning

- Interventions to promote recovery of motor control, sensation, and muscle functioning
- Interventions to promote recovery of lost functions e.g. arm and hand use, mobility, global functioning, self-care activities
- Interventions to optimise use of remaining functions using assistive devices e.g wheelchair, walking frames, chairs lift etc
- Interventions to reduce the effect of social restraints on the participation of the patient in the community

Interventions to promote restoration of body function and structure







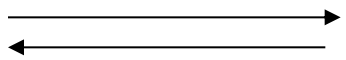






Definition of stroke

Rapid onset of clinical signs of focal or global disturbance of cerebral function of presumed vascular origin lasting more than 24 hours duration (WHO, 1989).

Stroke  Cerebrovascular accident

cognitive, motor, sensory, perceptual and language deficits

location, size of brain lesion, collateral blood flow and early mgx determine long lasting deficits

Key statistics

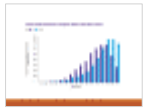
There are more than **100,000 strokes** in the UK each year. That is around one stroke every **five minutes**.

There are over **1.2 million stroke survivors** in the UK.

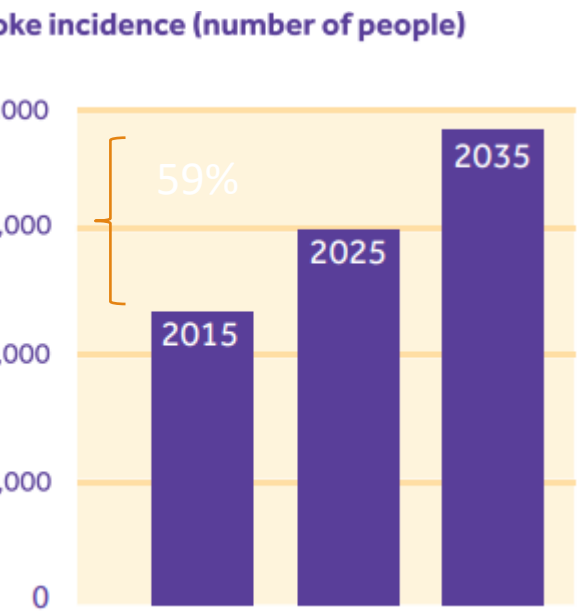
Every **two seconds**, someone in the world will have a **stroke**.

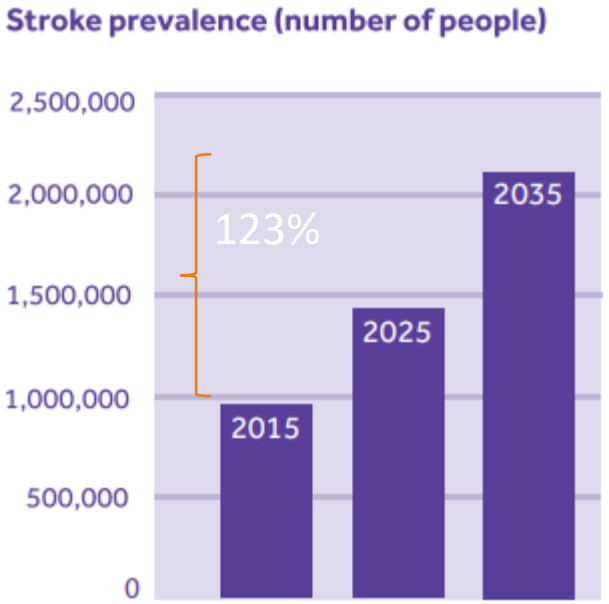
Stroke is the **fourth biggest killer in the** in the UK. Fourth in England and Wales, and the third biggest killer in Scotland and Northern Ireland.

More than **400** children have a **stroke** every year in the UK.

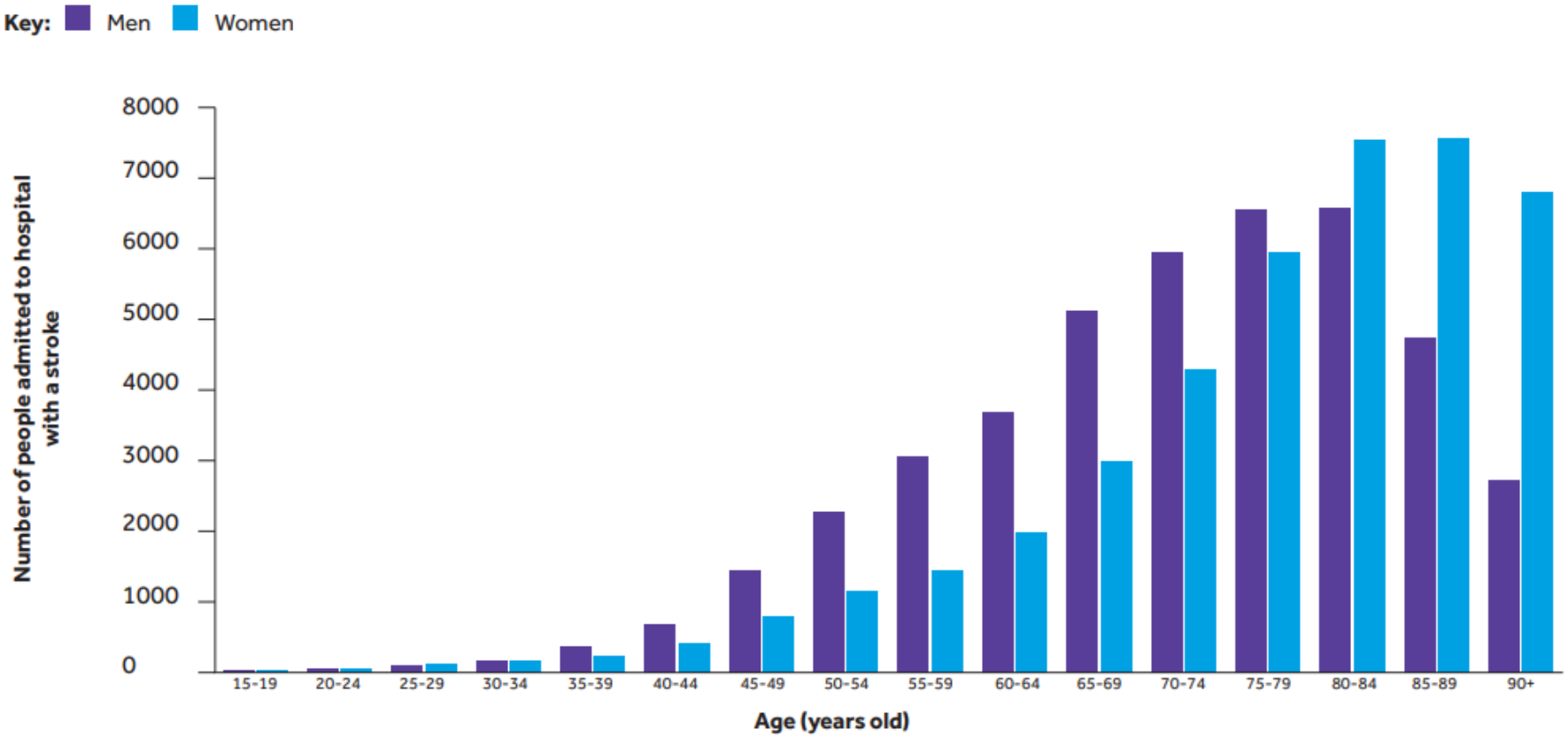


A **third** of stroke survivors experience depression after having a **stroke**.





Stroke Hospital Admissions in England, Wales and Northern Ireland [5]



How common is stroke in the UK?

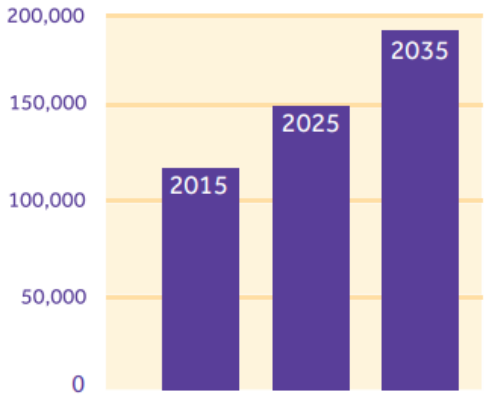
- There are more than 100,000* strokes in the UK each year. [5 6]
- That is around one stroke every five minutes in the UK.
- Between 1990 and 2010 the incidence of strokes fell by almost a quarter.[7]
- Around 1 in 6 men will have a stroke in their life.[8]
- Around 1 in 5 women will have a stroke in their life.[8]
- The rate of first time strokes in people aged 45 and over is expected to increase by 59% in the next 20 years (between now and 2035).[9]
- In the same period, it's estimated that the number of stroke survivors, aged 45 and over, living in the UK is expected to rise by 123%.[9]

*It's hard to determine with accuracy how many people have a stroke each year. We know that there are at least 100,000 strokes, but there may be more. The available data relies on hospital admissions and does not include who die before reaching hospital, or were not treated in hospital. For these reasons, we have agreed, together with the professional and academic community, to utilise this estimate.

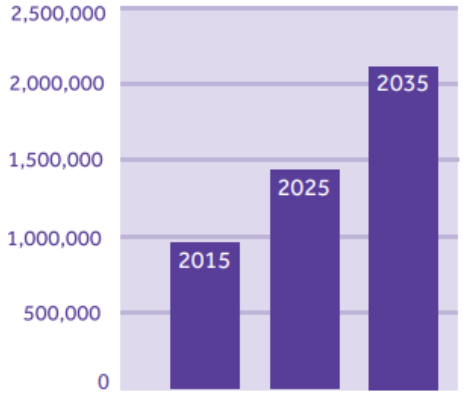
Recurrence

- Stroke survivors are at greatest risk of having another stroke in the first 30 days following a stroke.[10]
- Around 1 in 4 stroke survivors will experience another stroke within five years.
- In England, Wales and Northern Ireland, over a quarter of people who have a stroke have had a previous stroke or TIA.[5]
- In England, Wales and Northern Ireland 1 in 17 (6%) stroke patients have another stroke while still in hospital.[5]
- In Scotland the number of people having TIAs has increased by 17% in the last decade (since 2006).[11]

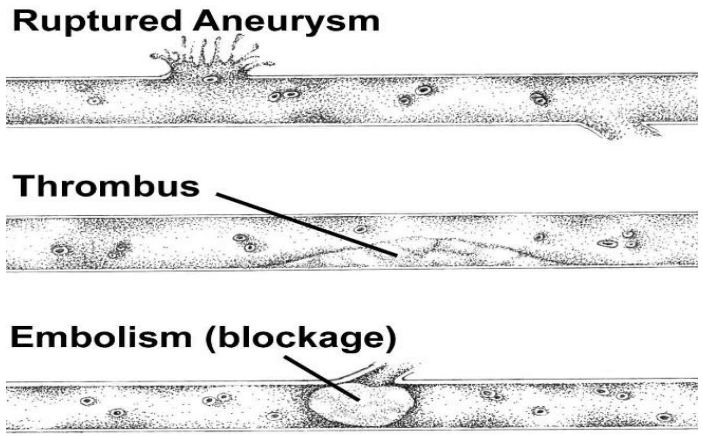
Stroke incidence (number of people)



Stroke prevalence (number of people)



Pathophysiology of stroke



Brain cell death
(ischaemic core + penumbra)

Cascade of
chemical factors
e.g. excitatory
neurotransmiter
(glutamate)

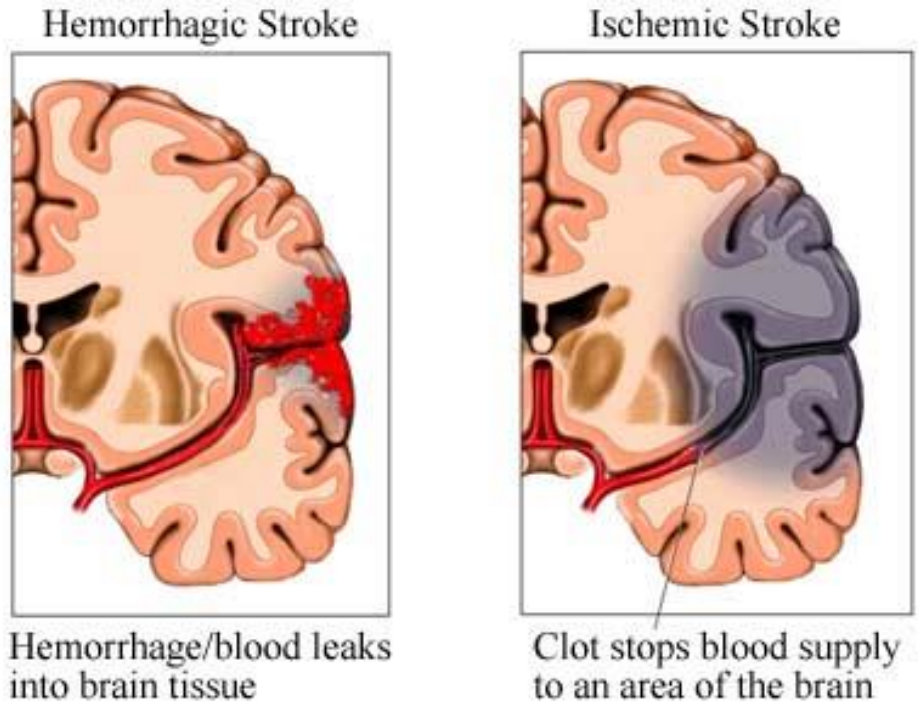
Are all strokes the same?

Ischaemic
- brain infraction caused by low CBF e.g. thrombus, embolism

Haemorrhagic
- bleeding from aneurysm or trauma
- Intracerebral (lipohyalinosis)

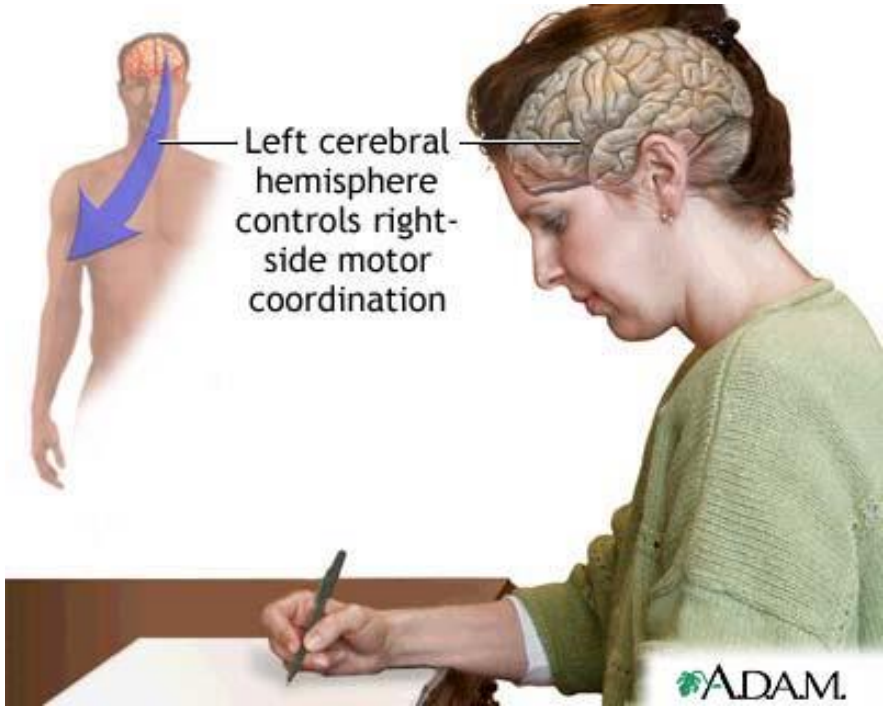
- Subarachnoid
  - bleeding into the subarachnoid space from aneurysms or arteriovenous anastomosis/malformations

Ischaemic is the commonest type (61-81%)



Side of paralysis





Common Signs and symptoms

Middle cerebral artery (MCA) syndromes

- Paresis of contralateral face, arm and leg (least)
- Sensory impairment over contralateral face, arm and leg (pain, temp, touch, vibration, position, two-point discrimination, stereognosis)
- Expressive (Motor speech) aphasia
- Receptive aphasia
- Perceptual problems such as unilateral neglect, apraxias, depth perception problems
- Ataxia of contralateral limb (sensory ataxia)

Other impairments

Dysphagia (inability to swallow) ↑aspiration risks, delayed swallowing (86%)

Perceptual dysfunction (unilateral neglect, agnosia, apraxia, Pusher syndrome)

Cognitive dysfunction (attention, memory or executive functions)

Affective disorders (emotional disorders, depression (1/3) (L) more)

Seizures (small %)

Bladder and bowel dysfunction (acute phase)

Indirect impairments

Venous Thromboembolism
- Immobility
- +ve Homan's sign

Skin breakdown
- Pressure sores

↓ Flexibility
- connective tissue
- atrophy
- contractures

Shoulder subluxation and Pain
- 70-80% of cases
- ↓proprioception
- Recurrent subluxation
- Adhesive capsulitis
- Shoulder trauma

Reflex Sympathetic Dystrophy
- Severe shoulder, arm and hand pain, hand oedema, Dystrophic changes in skin, bone and muscles

Functional Problems

Consequent upon the:

- Motor and perceptual impairments, sensory, disorientation, communication, and decreased cardiorespiratory endurance.

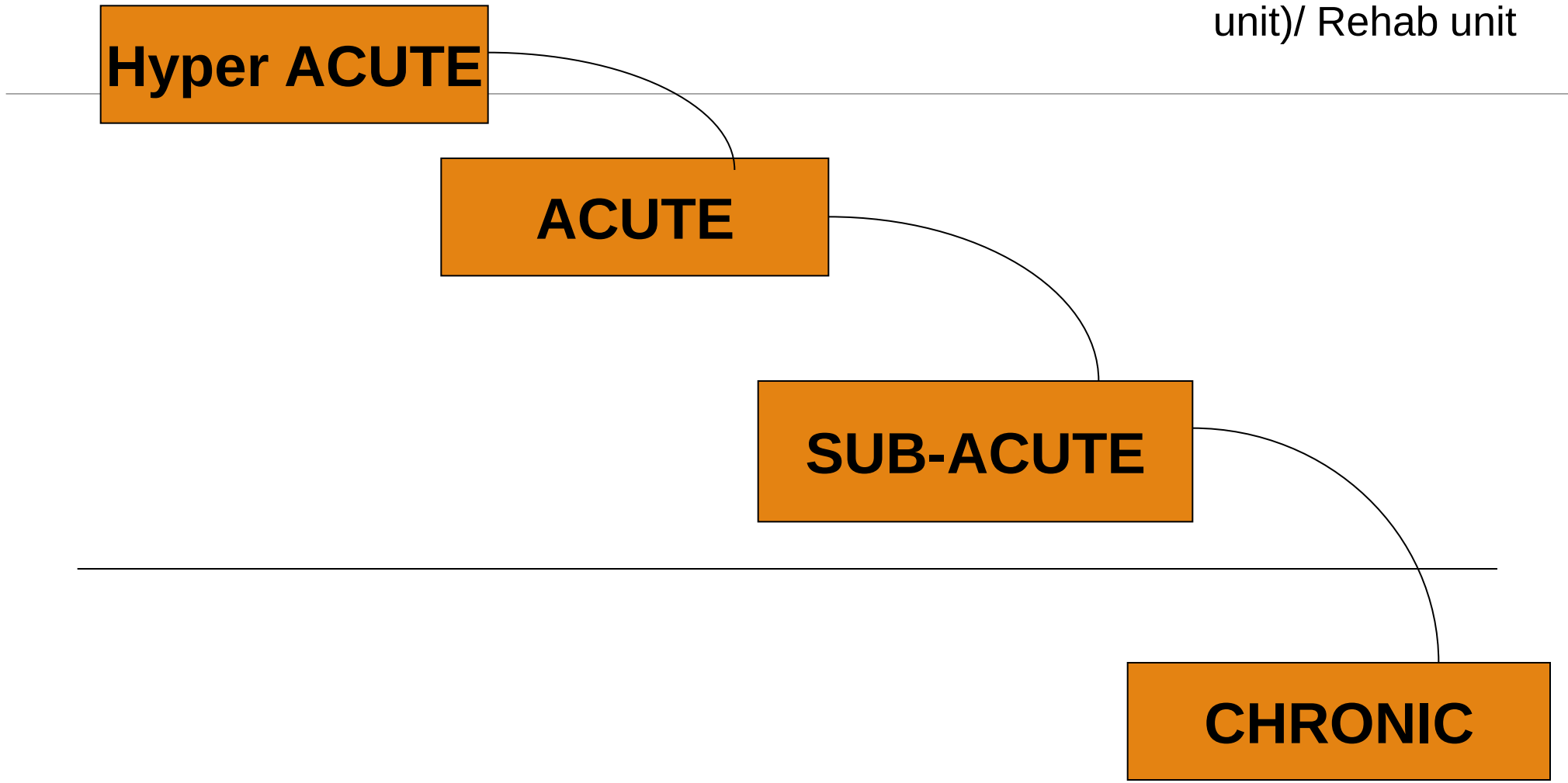
Basic ADL skills

- Toileting, bathing, grooming, self-feeding etc.

Mobility problems

- turning in bed, sitting, sit to stand, transfers, standing, walking, stairs and outdoor walking

Time since stroke



Chronic Case study

MT is a 55year old teacher who lives in a terraced house which has one flight of stairs with one of her two sons. She got divorced 10 years ago. She has a history of hypertension; she smokes and is overweight. Her hobbies are gardening, and twice a week she plays bridge at her local club.

She suffered a stroke, six months ago, when she woke up and was unable to get out of bed or talk properly. She was taken to hospital and on admission she was paralysed on the right side of her body with low tone and reduced reflexes and could not speak. Patient was treated in the hospital and discharged after 3 months.

At 6 months, MT is able to get up and downstairs with help from her son. She can also walk outside her house. However, MT has fallen three times in the past 3 months, twice while walking outside the house and once while getting out of bed at night. She continues to receive community physiotherapy.

Physiotherapy Treatment plan

Issues	Plan
Multiple falls	Assessment of general, motor sensory and cognitive assessment to understand the cause of the fall. Dual task was implicated and forgot to switch on the light on one occasion
	Treatment was focused on balance rehabilitation with dual task as an element.
	Getting up from floor after falling
	Self-management exercises focused on improving balance exercises

Self-Directed Study

Learn and understand key neurological terminologies.

Read Glossary of neurological terms:

Kenyon K & Kenyon J (2018) The Physiotherapist's pocketbook: essential facts at your fingertips. Page 105 -109

Link here: https://r1.vlereader.com/EpubReader?ean=1780702077982# You may have to log into the reading list to access this.

For future reference with regards to placement:

Common Neurological Terms for placement – Weblearn.